Module 18

Entity-Relationship Model/1

Milav Dabgar

www.milav.in

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Module Recap

- Predicate Calculus

Module Recap

- Predicate Calculus
- Tuple Relational and Domain Relational Calculus

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- Predicate Calculus
- Tuple Relational and Domain Relational Calculus
- Equivalence of Relational Algebra and Relational Calculus

Module Objectives

Goals

- To understand the Design Process for Database Systems

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- To understand the Design Process for Database Systems
- To study the E-R Model for real world representation

Module Outline

- Design Process

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- Design Process
- E-R Model

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- Design Process
- E-R Model
  - Entity and Entity Set

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- Design Process
- E-R Model
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  - Relationship
  - Cardinality
  - Attributes
  - Weak Entity Sets

Design Process

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- Conforms to limitations of the target medium
- Meets implicit or explicit requirements on performance and resource usage
- Satisfies implicit or explicit design criteria on the form of the artifact
- Satisfies restrictions on the design process itself, such as its length or cost, or the tools available for doing the design

Abstraction

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- Abstraction provides the major tool to handle Disorganized Complexity by chunking information
- Ignore inessential details, deal only with the generalized, idealized model of the world

Models

Model Building

- Physics: Time-Distance Equation, Quantum Mechanics

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- Build new models upon old proven models

Design Approach

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  - **Computer Science Decision**: What relation schema should we have and how should the attributes be distributed among the various relation schema?
- **Physical Model**: Deciding on the physical layout of the database

Design Approach (3): Database Designing: Logical Model

- **Entity Relationship Model**

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- **Database Normalization**
  - Formalize what designs are bad, and test for them

ER Model

Entity Relationship (ER) Model

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- The ER model also has an associated diagrammatic representation, the ER diagram, which can express the overall logical structure of a database graphically

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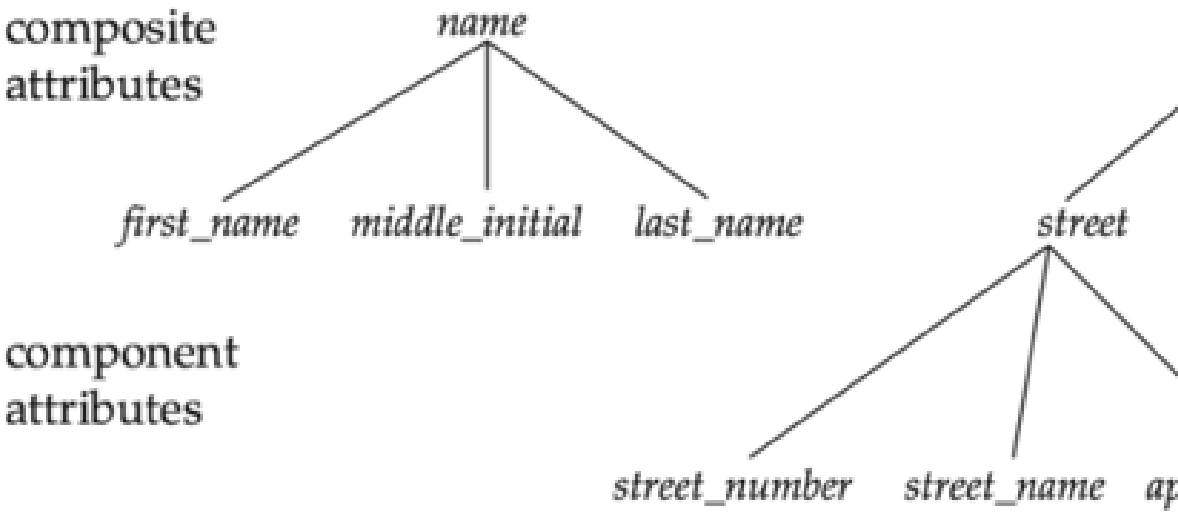
- Domain: Set of permitted values for each attribute

Attributes (2): Composite



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  - Primary key of an entity set is represented by underlining it.

Entity Sets instructor and student

76766	Crick
45565	Katz
10101	Srinivasan
98345	Kim
76543	Singh

98988	Tanaka
12345	Shankar
00128	Zhang
76543	Brown
76653	Aoi
23121	Chavez

Relationship

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- A relationship set is a mathematical relation among $n \geq 2$ entities, each taken from entity sets

$$\{(e_1, e_2, \ldots e_n) \mid e_1 \in E_1, e_2 \in E_2, \ldots, e_n \in E_n\}$$

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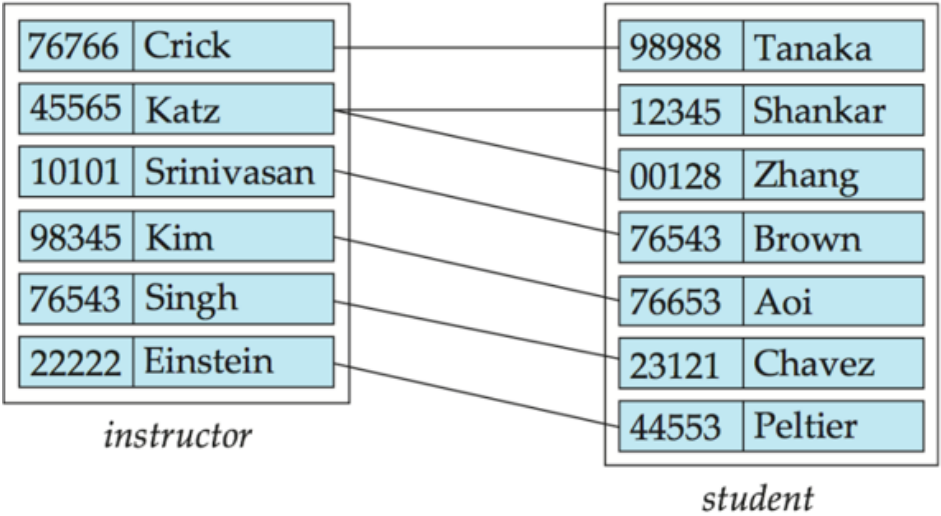
- Example: $(44553, 22222) \in$ advisor

Relationship Set (2) advisor



Relationship Sets (3)

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- For instance, the `advisor` relationship set between entity sets `instructor` and `student` may have the attribute `date` which tracks when the student started being associated with the advisor

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Relationship Set (4): Degree

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- Relationships between more than two entity sets are rare. Most relationships are binary
  - Example: students work on research projects under the guidance of an instructor.
  - relationship `proj_guide` is a ternary relationship between `instructor`, `student`, and `project`

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- BUT: When converting back to tables, in some cases the attribute gets reintroduced, as we will see later

Cardinality

Mapping Cardinality Constraints

Overview

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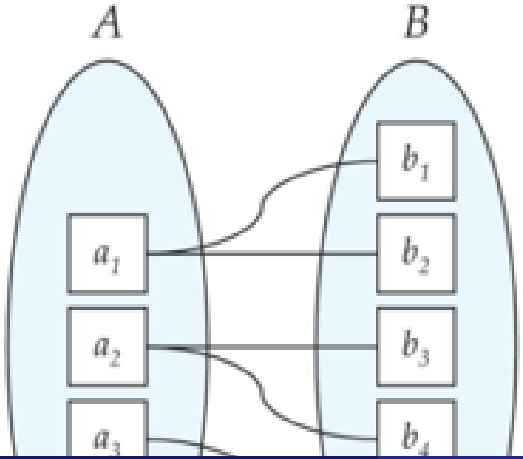
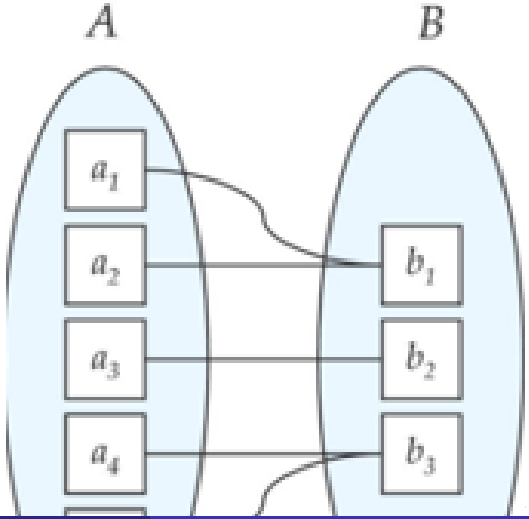
Note: Some elements in A and B may not be mapped to any elements in the other set





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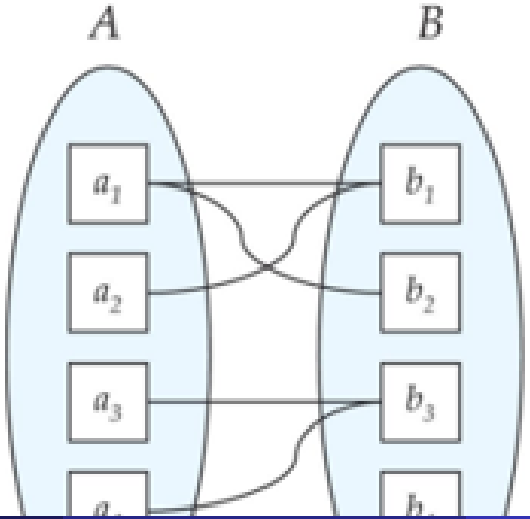

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  - A weak entity set is an entity set that does not contain sufficient attributes to uniquely identify its entities

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  - Primary Key of Weak Entity Set = Its own discriminator + Primary Key of Strong Entity Set
- Weak entity set must have total participation in the identifying relationship. That is all its entities must feature in the relationship

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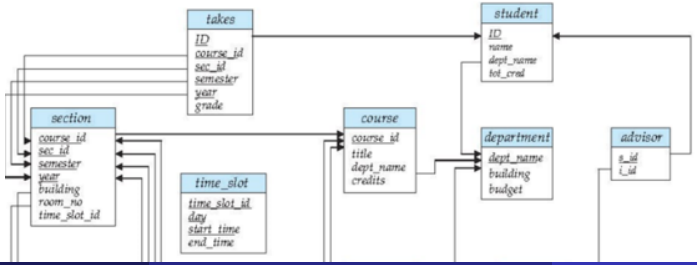
- By total participation in BA, each apartment must be present in at least one building
- In contrast, Building has partial participation in BA only as there might exist some buildings which has no apartment
- Primary Key: To uniquely identify any apartment
  - First, building no is required to identify the particular building
  - Second, door no of the apartment is required to uniquely identify the apartment
  - Primary key of Apartment = Primary key of Building + Its own discriminator = building no + door no

Weak Entity Sets (4): Example

Example

- Consider a `section` entity, which is uniquely identified by a `course_id`, `semester`, `year`, and `sec_id`.



Weak Entity Sets (4): Example

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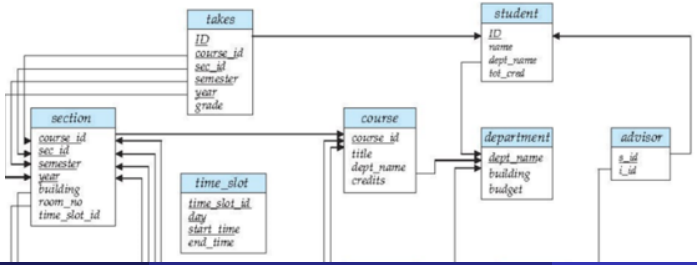

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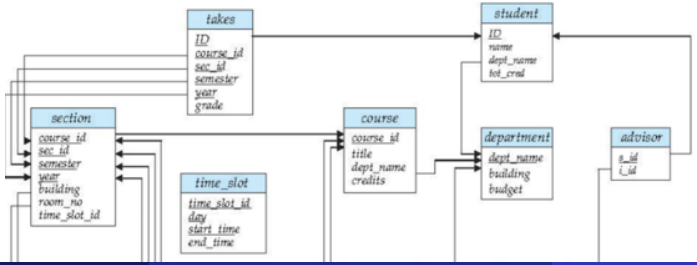
- Note that the information in `sec_course` is redundant, since `section` already has an attribute `course_id`, which identifies the course with which the section is related.



Module Summary

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- Introduced the Design Process for Database Systems

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- Introduced the Design Process for Database Systems
- Elucidated the E-R Model for real world representation with entities, entity sets, attributes, and relationships

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